toys10_08-band-racer

Rubber-band Racer: manufacturing manual

> **Status: design only; nothing has been cut or assembled.**

1. What is made

1. Build the laser-cut chassis (no tools), wind the propeller-free rear axle with a thumb wheel, place on the floor and release.
2. Race on a 5 m track; try 2 or 3 bands, different wheels.

2. Files and materials

- `toys10/out/racer_parts.svg` and `.dxf`: **provisional first-pass drawing** generated by `toys10/sims/gen_cut_files.py` (150 x 60 mm chassis, four 50 mm wheels with 46 mm groove discs, thumb wheel): never cut; distance estimates in `toys10/docs/results_racer.md` come from assumed constants.
- Material: **3 mm MDF**; layout/sheet: 300 x 200 mm.

Cost per set (about 2,000 units, estimate):

Item	Rs
Laser-cut chassis + wheels	45
Axles + O-rings	18
Rubber bands ×6	6
Box + instructions	30
Assembly + packing	20

3. Making it

Follow `manuals/STANDARD_PROCEDURES.md` section H (laser cutting) for kerf tests and finishing, then:

1. Cut a cardboard version first; check axles turn freely on 4 mm dowels in 4.2 mm holes.
2. Cut MDF parts; glue each wheel from two 50 mm discs with one 46 mm disc between them to form the groove for the O-ring tyre.
3. Assemble chassis, axles, wheels, hook for the rubber band, thumb wheel on the rear axle.
4. Fit rubber bands (2 or 3) and O-rings; wind and release on a 5 m track.

4. Inspection (sample per batch)

- Runs 5-12 m from a full wind (the target; the model is an estimate): measure it, then correct the constants.
- Wheels do not wobble; band does not catch on the chassis.

Record on `manuals/templates/eol_test_sheet.md`.

5. Packaging and compliance

Toys QCO / IS 9873 applicability and testing; age grade and small-parts warnings; Legal Metrology labelling; paint/ink heavy-metal report; no licensed names or logos.

Add the age grade, small-parts warning, Legal Metrology marks and a printed instruction leaflet.

6. Risks and limits

- Distance model is unverified.
- Thin margin at Rs 449: sell as a 3-kit pack.

Cheap imported pull-back cars; low price point means thin margin unless sold as 3-kit packs.

`docs/results_racer.md`: distance model. **My first constant was absurd (60 m); it was cut by 6×.** All results are estimates from assumed constants (5–12 m target) and must be measured with the first prototype.